IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion

Executive Summary

Summary of methodologies

- Data collection
  - Web scraping
  - SpaceX API
- Data Cleaning
  - Pandas Dataframe
- Exploratory Data Analysis
  - Visualizations with Plots, Maps, and Dashboards
  - Exploration of data with SQL
- Model Building and Training
  - Logistic Regression, SVM, decision tree and KNN
- Model Evaluration
  - Confusion matrix
  - Accuracy on testing data

Summary of all results

- Identified key factors for landing success
- SVM prediction model performed best (84.8 % accuracy)
- Low amount of false negative predictions

Introduction

To compete with SpaceX relatively cheap rocket launches the successful landing of the first stage of Falcon 9 rockets is key to cutting the costs. The first stage is the biggest part of the rocket and it can be reused for future rocket launches if it lands successfully. Using the data from SpaceX, factors influencing the landing outcome are identified and predictions of landing outcomes are carried out using different machine learning models.

Accurately predicting the landing outcome of the first stage will improve cost prediction and can help in competing with SpaceX.

Section 1

Methodology



Methodology

Executive Summary

Data collection methodology:

Data was collected from the SpaceX API as well through web scraping and transformed into a Pandas data frame.

SpaceX API

From the SpaceX API information about Booster Version, Launch Site, Payload, Core and past Launches was gathered as a .json and normalized

The data was combined in a dictionary and transformed to a pandas data frame

Only Falcon 9 Booster Versions were used

Web scraping

- Information about Falcon 9 launches was gathered through webscraping from Wikipedia
- The html response was was used to create a BeautifulSoup object
- A data frame was created by parsing the launch html tables

Methodology

Data wrangling

- Missing numerical values were replaced with the mean
- The landing outcome was classified for successful (1) and not successful (0) landings

Exploratory data analysis (EDA) using visualization and SQL

- Visualization of success rates, launch sites, orbit types and payload with seaborn

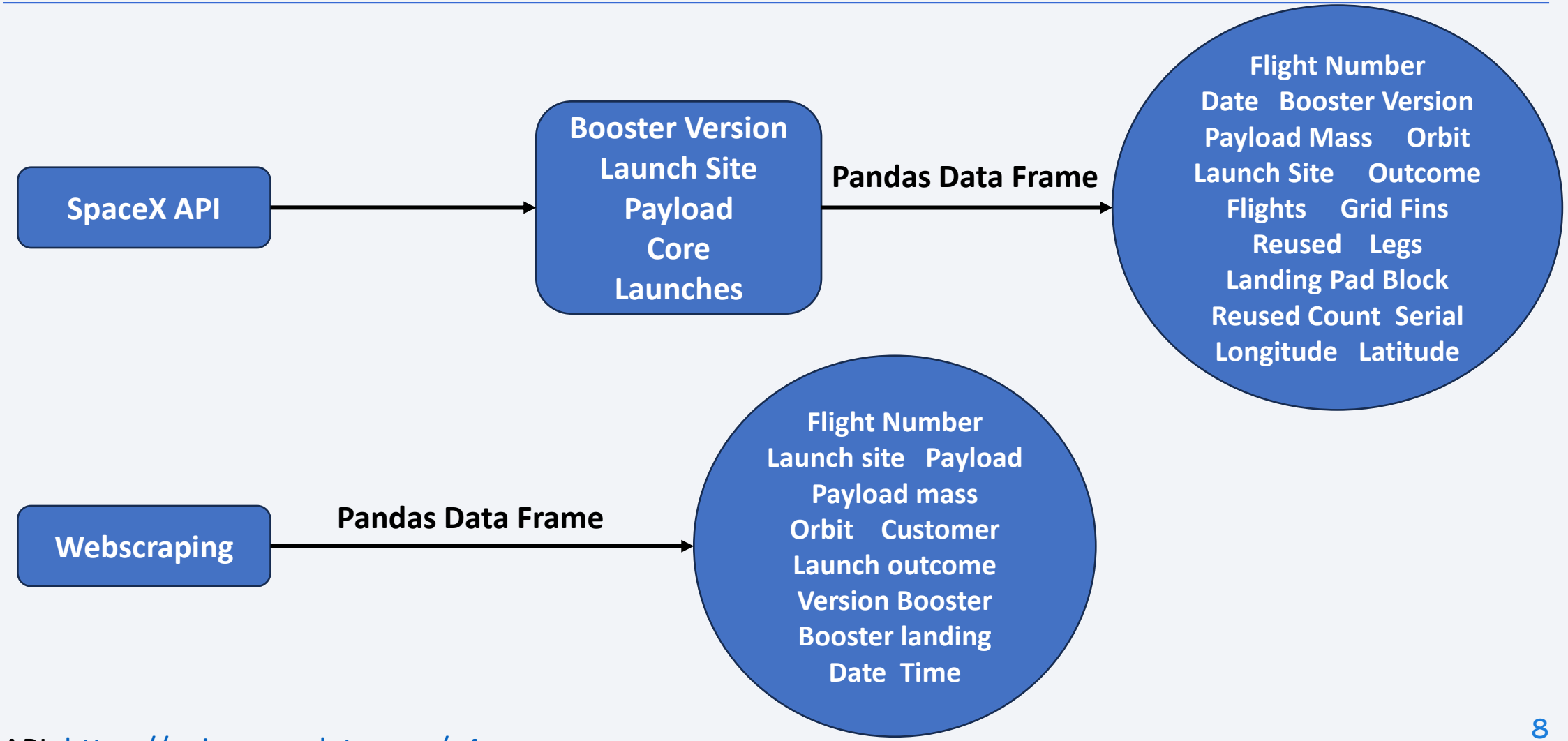
Interactive visual analytics using Folium and Plotly Dash

- Launch sites, successful and failed launces were marked on a map with folium and distances to the launch sites were calculated
- An interactive dashboard for visual analytics on SpaceX data was created

Predictive analysis using classification models

- How to build, tune, evaluate classification models
- A numpy Array was created from the “Class” column and the numerical values were standardized
- The data was split into training and testing data sets
- Logistic regression, SVM, decision tree and KNN was used and evaluated by scores and confusion matrices

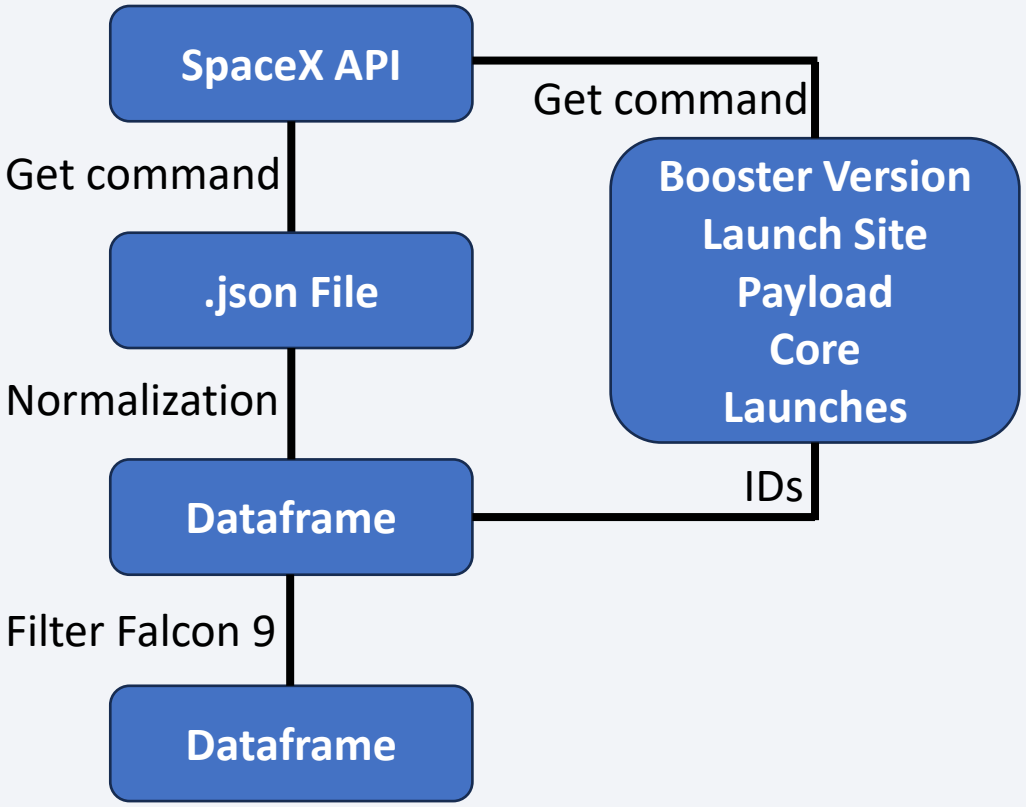
Data Collection



SpaceX API: https://api.spacexdata.com/v4
Webscraping: https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches

Data Collection – SpaceX API

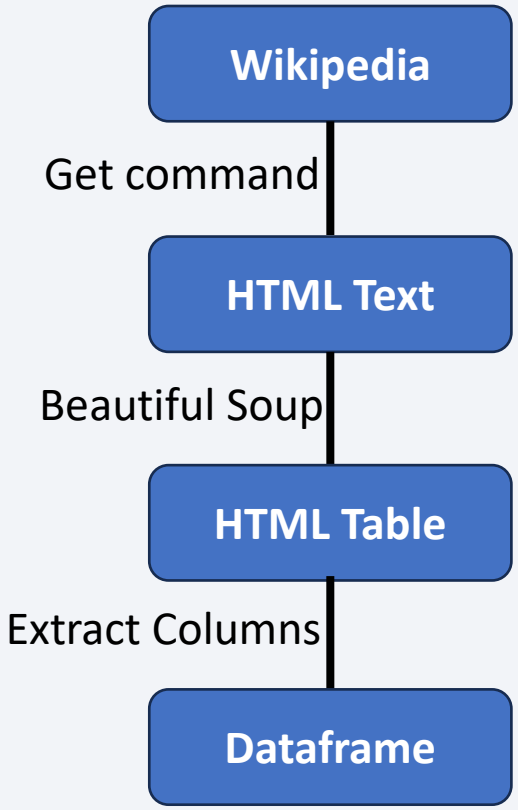
- Request and parse the SpaceX launch data using the GET request
- Data collection with SpaceX REST calls
- Filter the dataframe to only include Falcon 9 launches



https://github.com/hannesresearch/CourseraCode/blob/0e35266d7e45fd5ee28a088991e903c2e3721f00/Applied%20Data%20Science%20Capstone/jupyter-labs-spacex-data-collection-api.ipynb

Data Collection - Scraping

- Request the Falcon9 Launch Wiki page from its URL
- Extract all column/variable names from the HTML table header
- Create a data frame by parsing the launch HTML tables



https://github.com/hannesresearch/CourseraCode/blob/0e35266d7e45fd5ee28a088991e903c2e3721f00/Applied%20Data%20Science%20Capstone/jupyter-labs-webscraping.ipynb

Data Wrangling

- Dealing with Missing numerical Values by replacing them with the mean
- The number of launches on each site was calculated
- The number and occurrence of each orbit was calculated
- The number and occurrence of mission outcome was calculated
- A landing outcome label was created: 0 for unsuccessful landings and 1 for successful landings

https://github.com/hannesresearch/CourseraCode/blob/0e35266d7e45fd5ee28a088991e903c2e3721f00/Applied%20Data%20Science%20Capstone/labs-jupyter-spacex-Data%20wrangling.ipynb

EDA with Data Visualization

- Scatter plot of Flight Number and Launch Site to visualize the change in success rate for Launch Sites
- Scatter point chart of Payload Mass and Launch Site to visualize success rate depending on Payload Mass and Launch Site
- Bar chart showing success rate of each orbit type to check for relationships visually
- Scatter point chart of FlightNumber and Orbit type to see the change of success rate for different Orbit types
- Scatter point chart of Payload Mass and Orbit type to visualize success rate depending on Payload Mass and Orbit type
- Launch success yearly trend as line chart

https://github.com/hannesresearch/CourseraCode/blob/fc7386d5310841a29a8563a18e9cf3af4c917e21/Applied%20Data%20Science%20Capstone/edadataviz.ipynb

EDA with SQL

- Filtered data by unique values and specific conditions
- Calculated total values, averages etc.
- Sorted values by value counts
- Extracted month names from dates
- Used nested queries for more complex requests

https://github.com/hannesresearch/CourseraCode/blob/fc7386d5310841a29a8563a18e9cf3af4c917e21/Applied%20Data%20Science%20Capstone/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

- All launch sites were marked with circles on the map, to visualize an operational zone of the launch sites
- Successful and failed launches were marked on the map as clusters using colored markers for successful/failed launches. This reduces clutter and visibility, while conserving maximum information
- Proximities of launch sites were calculated and visualized on the map as lines to see distances to railways, highways, coastline and cities

https://github.com/hannesresearch/CourseraCode/blob/fc7386d5310841a29a8563a18e9cf3af4c917e21/Applied%20Data%20Science%20Capstone/lab_jupyter_launch_site_location.ipynb

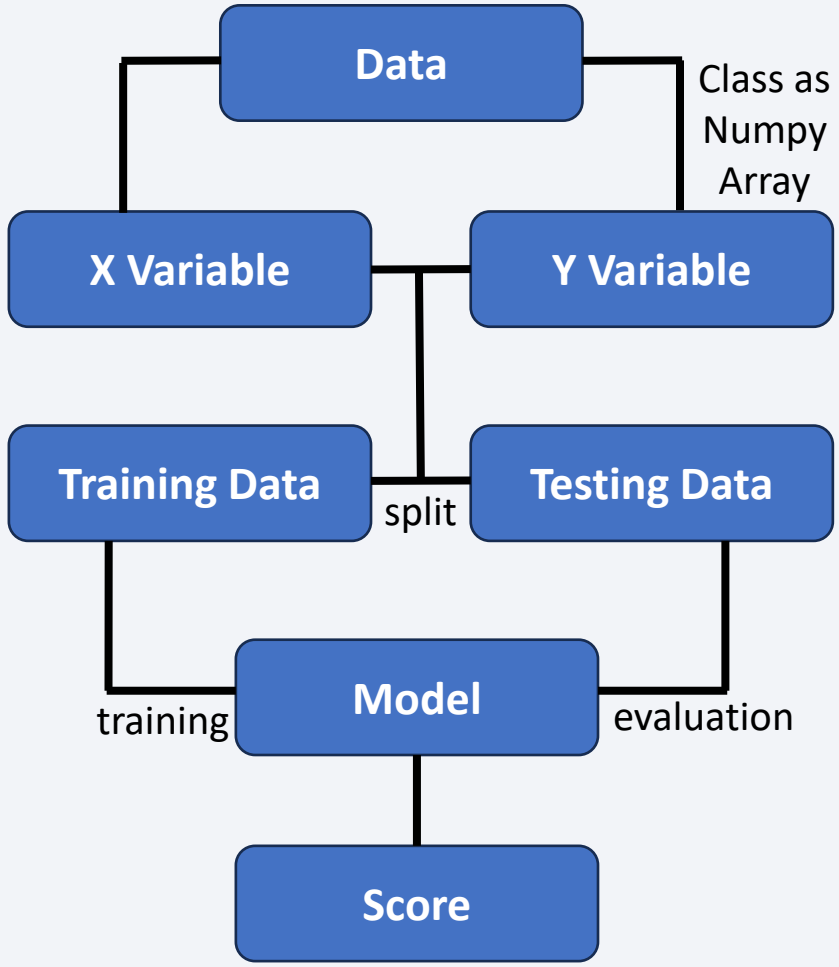
Build a Dashboard with Plotly Dash

- A drop-down Input Component for launch sites was added to select launch sites independantly
- A pie-chart was added based on launch site selected, showing the success rates
- Selection of all launch sites shows the total success launches by site
- A range slider for payload was added to filter by payload mass
- A scatter chart was added showing success rate depending on payload, showing the influence of payload to the success rate

https://github.com/hannesresearch/CourseraCode/blob/fc7386d5310841a29a8563a18e9cf3af4c917e21/Applied%20Data%20Science%20Capstone/spacex-dash-app.py

Predictive Analysis (Classification)

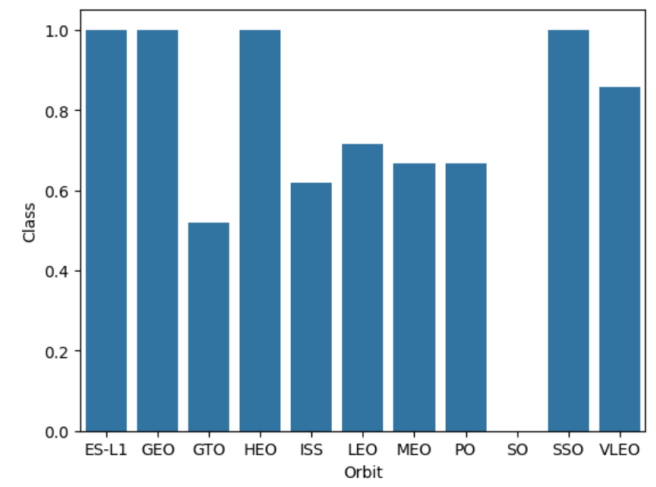
- Summarize how you built, evaluated, improved, and found the best performing classification model
- You need present your model development process using key phrases and flowchart
- The “Class” column was used as Numpy Array and assigned to the Y variable
- The rest of the data was standardized and used as variable X
- The data was split into test and training data sets
- All classifications models were trained on the training set and evaluated on the test set by accuracy and a confusion matrix
- All classification models were optimized using a set of parameters with GridSearchCV
- Logistic Regression, SVM, decision tree and KNN were trained, with SVM reaching the highest accuracy

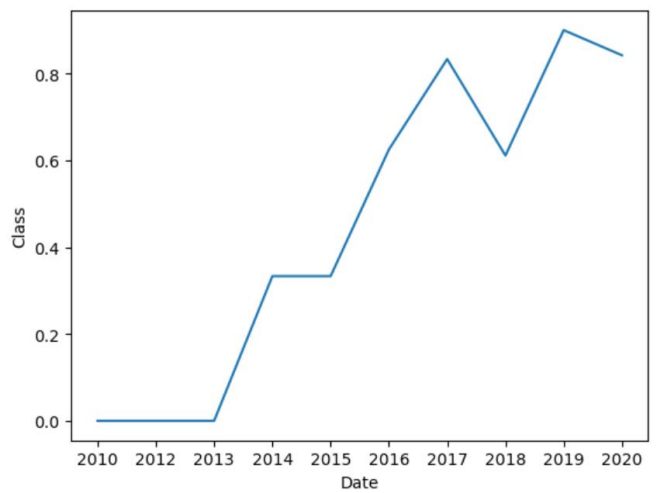


https://github.com/hannesresearch/CourseraCode/blob/fc7386d5310841a29a8563a18e9cf3af4c917e21/Applied%20Data%20Science%20Capstone/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

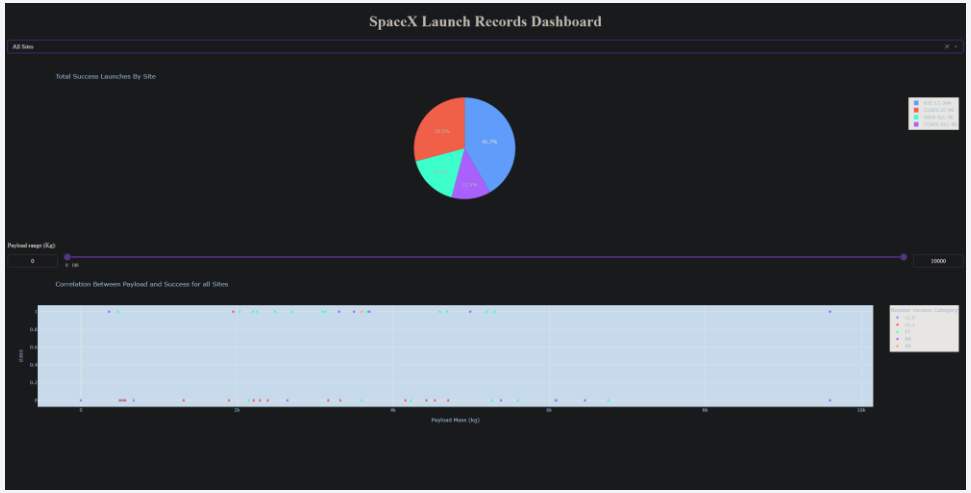
Results

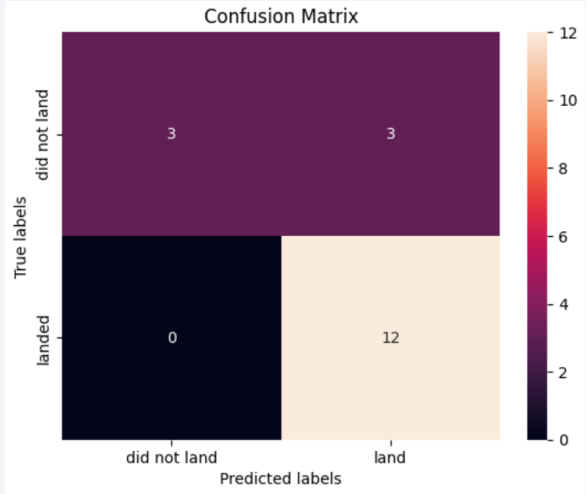
Exploratory data analysis results





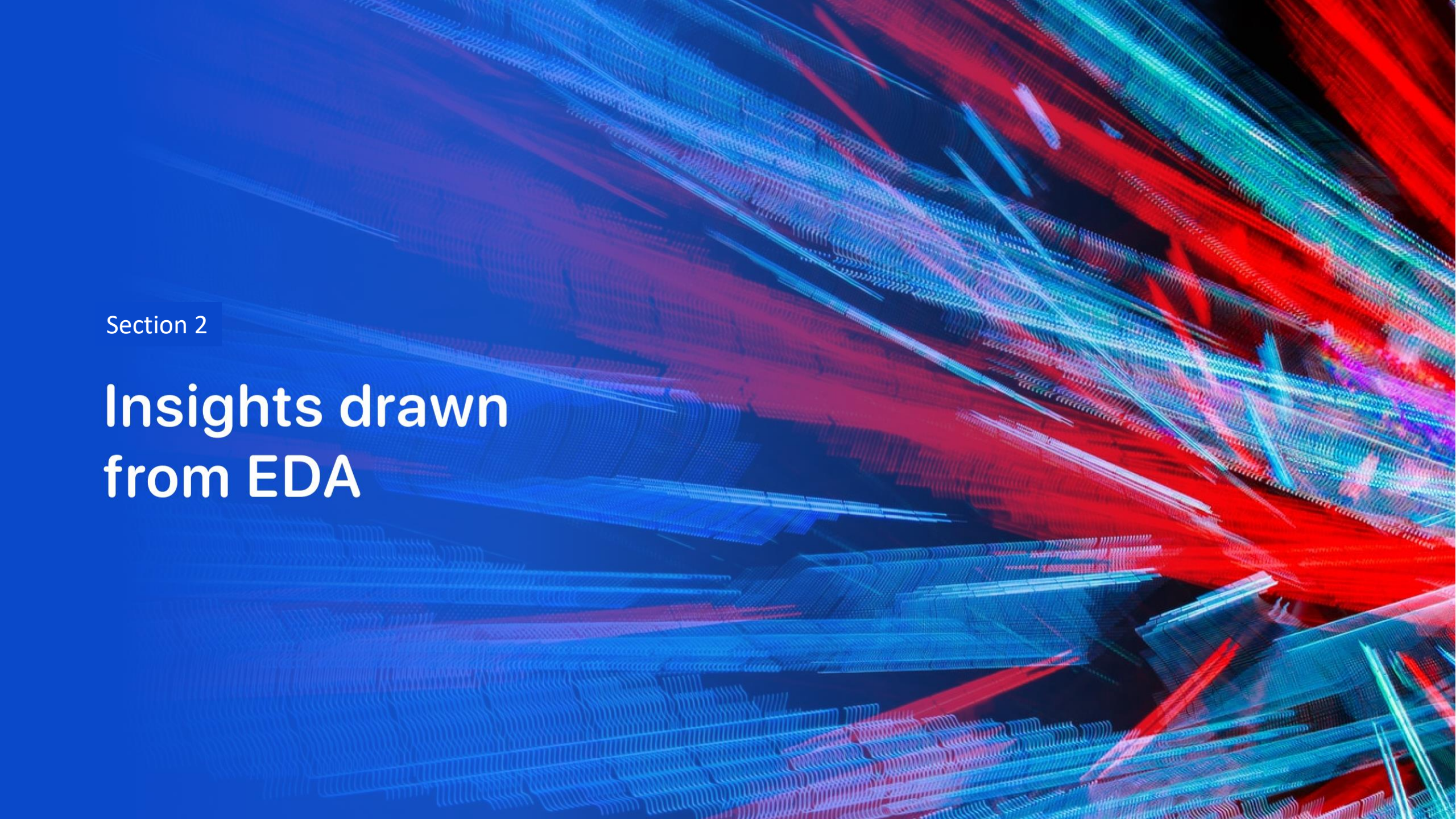
Interactive analytics demo





Predictive analysis results

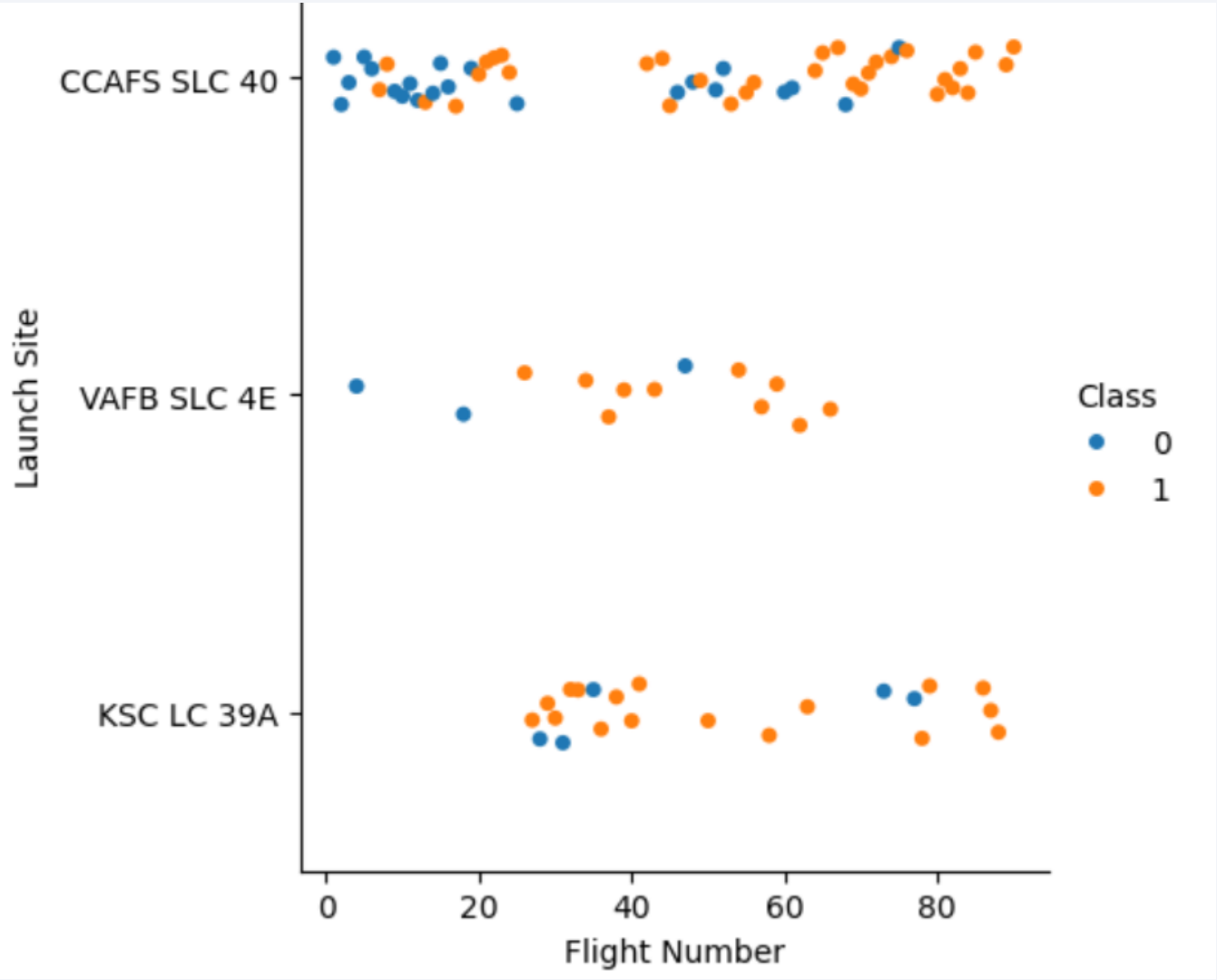
Accuracy SVM 84,8 %


Section 2
Insights drawn from EDA

Flight Number vs. Launch Site

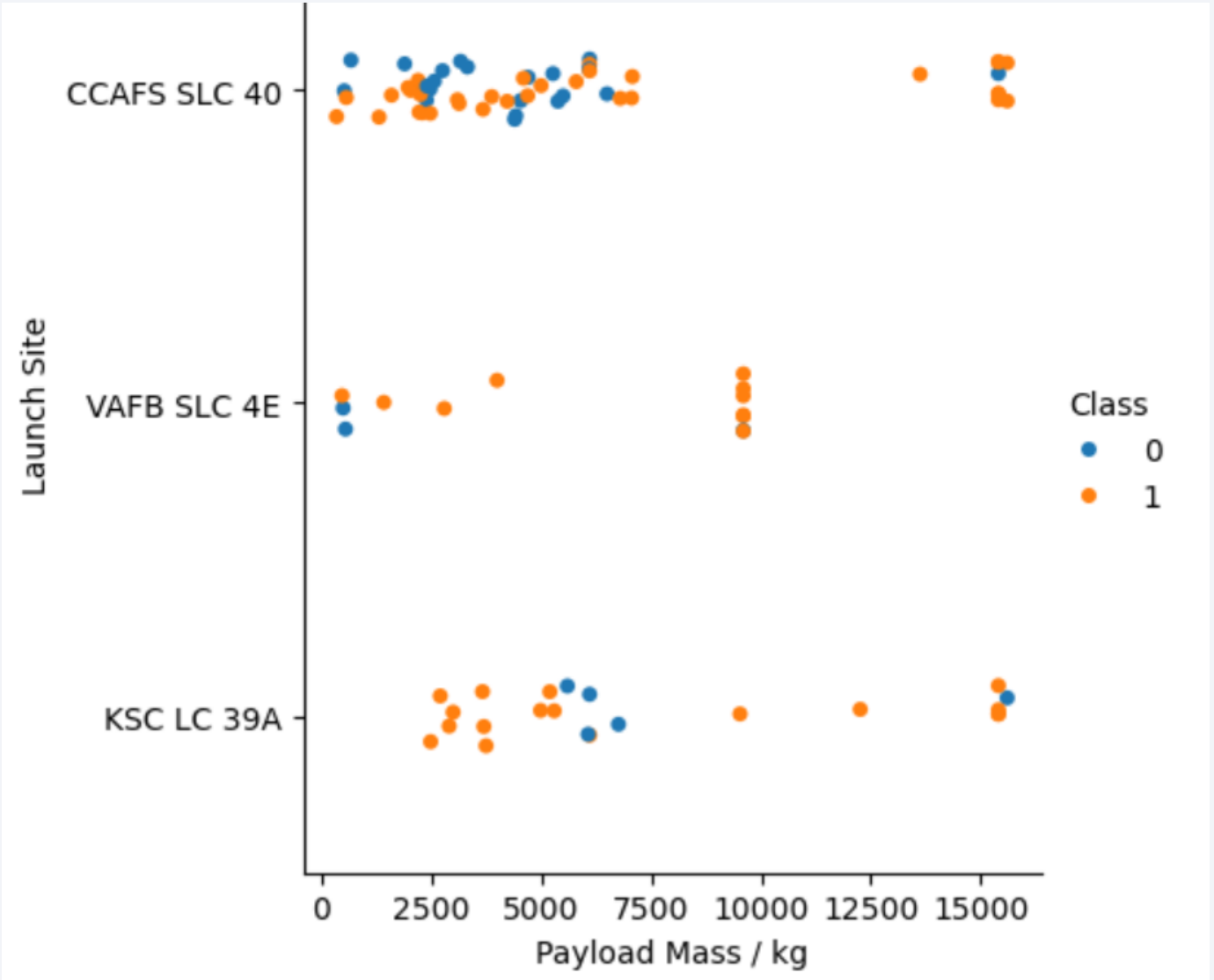
Flight Numbers for the 3 different launch sites are shown. It can be seen that CCAFS SLC 40 has the greatest number of flights, and the success rate increases with the flight number

KSC LC 39A however has a very high success rate while maintaining a high number of flights



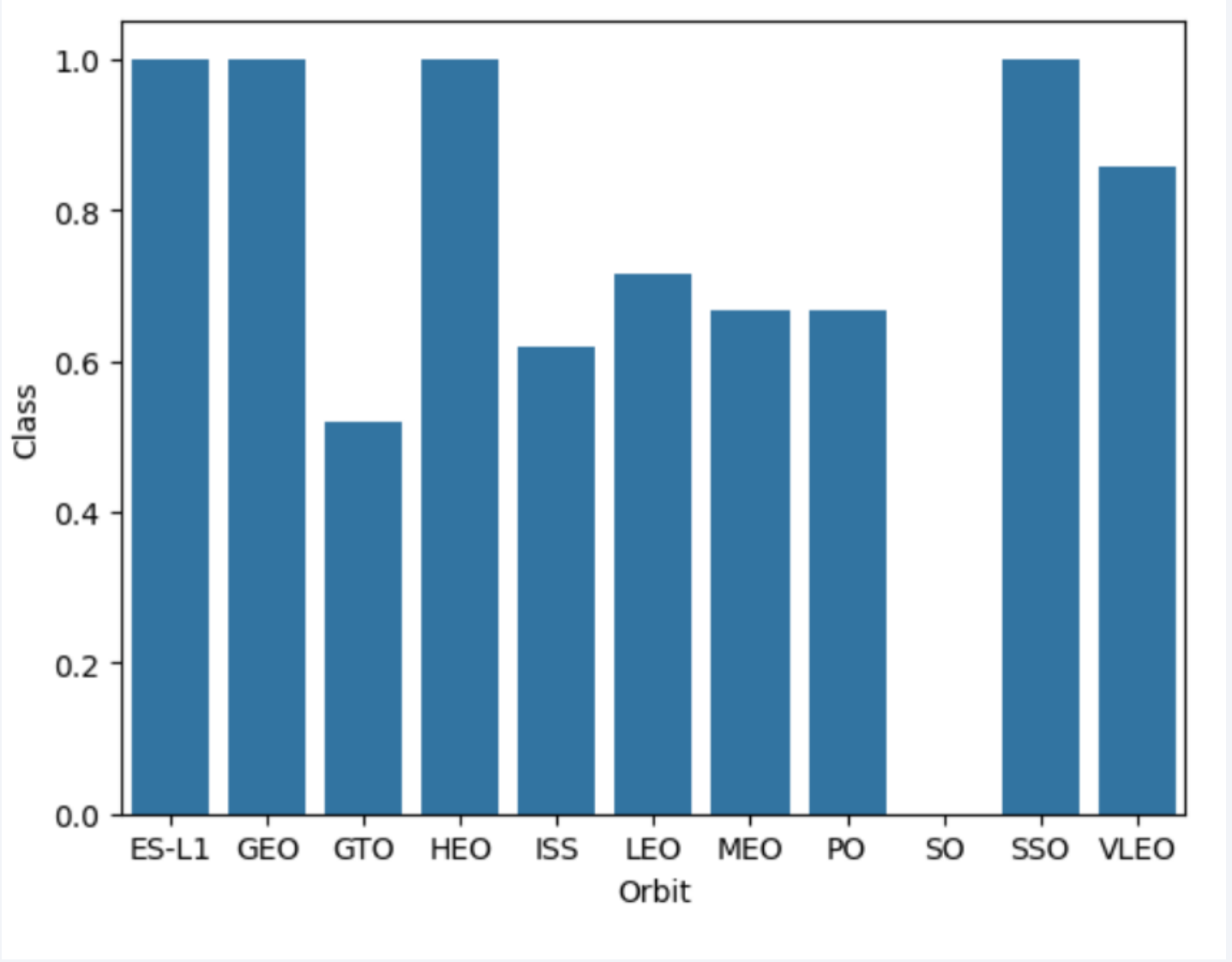
Payload vs. Launch Site

The payload of the different launch sites is shown. It can be seen that high payloads are not used at VAFB SLC 4E and most failure landings are at lower payloads.



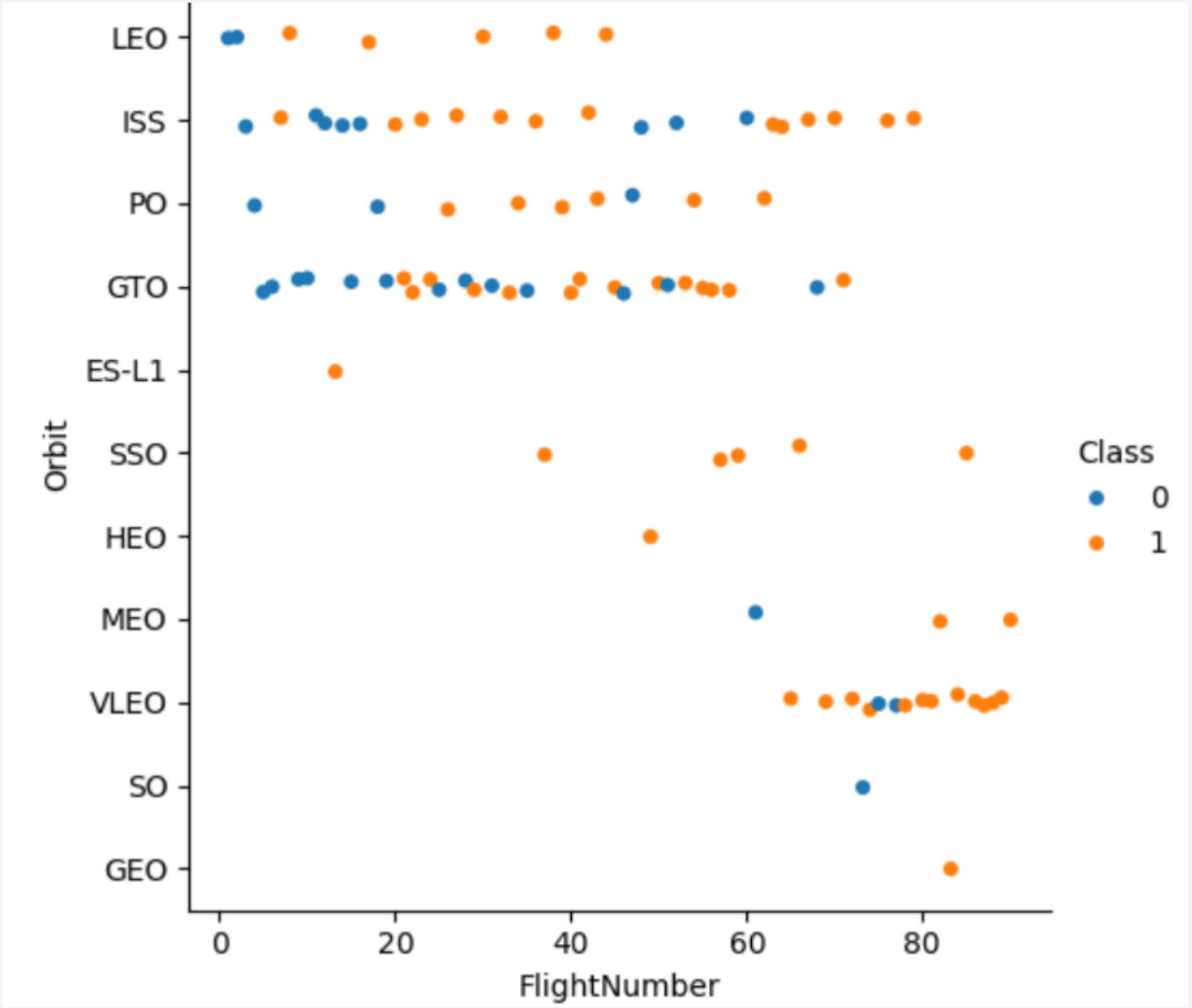
Success Rate vs. Orbit Type

Success rates are highest for ES-L1, GEO, HEO and SSO orbit types and lowest for GTO, ISS and SO.



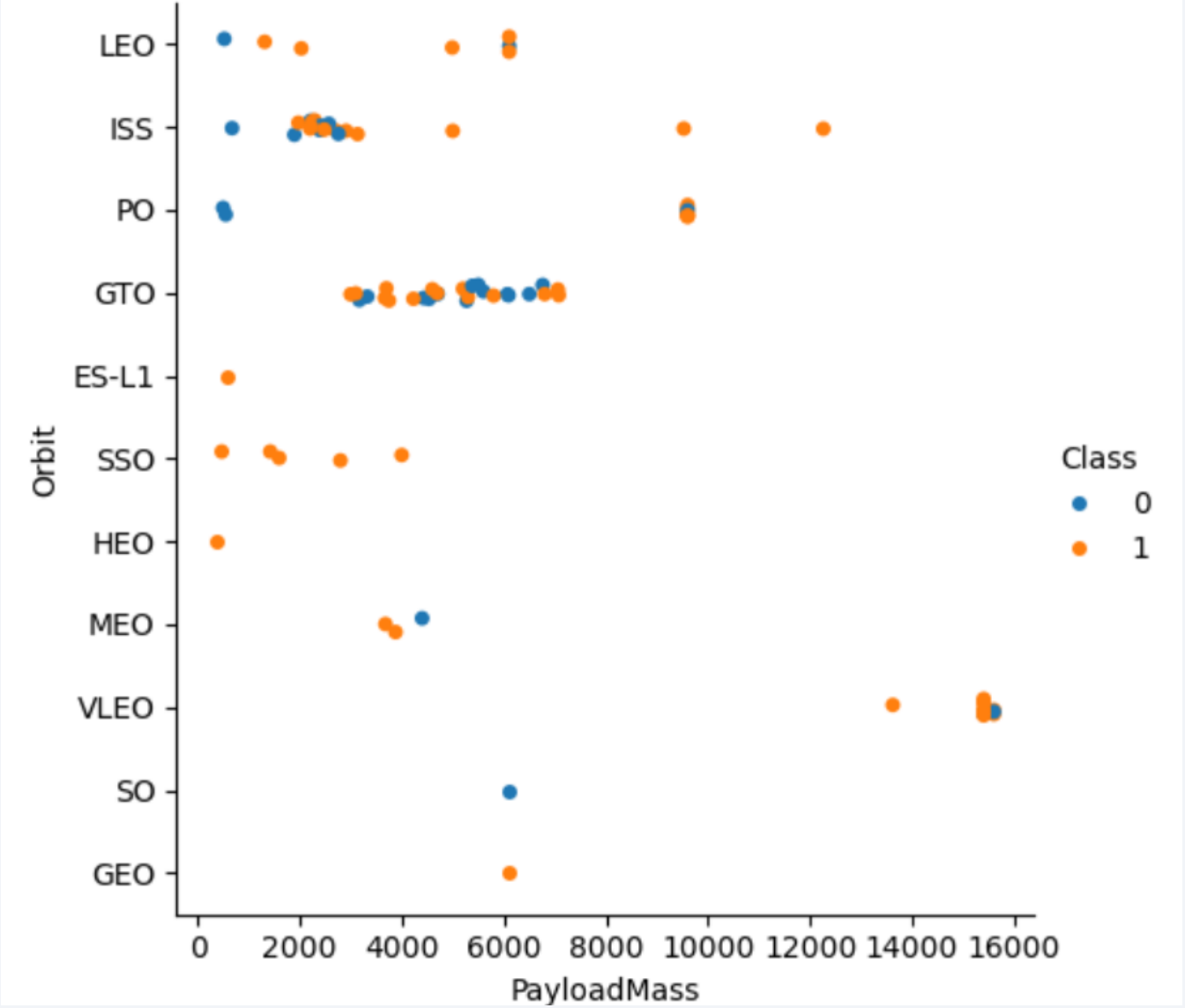
Flight Number vs. Orbit Type

- It can be seen that ES-L1, HEO, SO and GEO only had one flight. The most recent flights were mainly in VLEO and mostly successful.



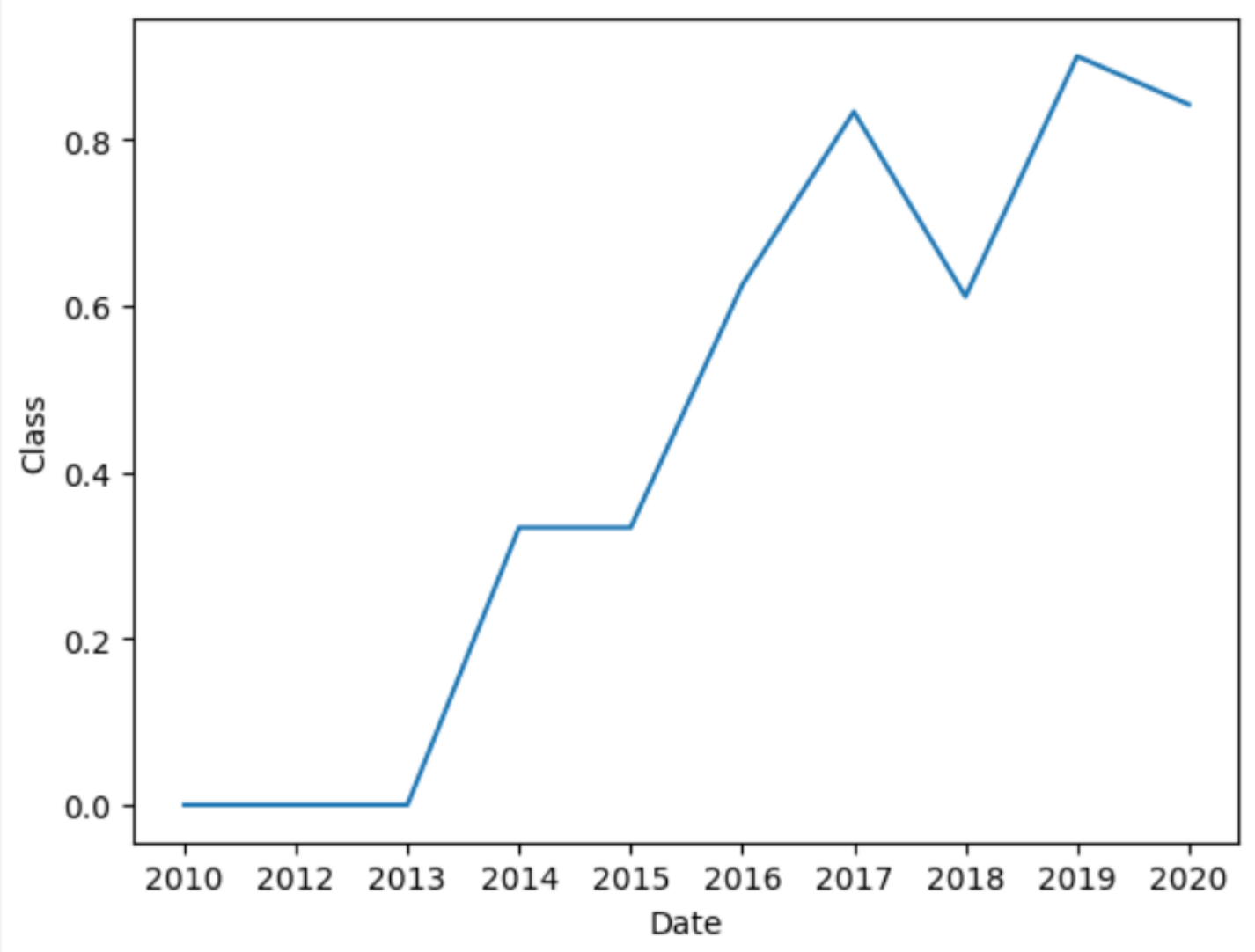
Payload vs. Orbit Type

Payloads over 10000 kg are only in ISS, PO and mainly in VLEO. The other orbit types are in the range of 1000 - 8000 kg.



Launch Success Yearly Trend

The average success rate starts very low and increases from 2013 to 2017 to over 80%.



All Launch Site Names

%sql SELECT DISTINCT Launch_Site FROM SPACEXTABLE;

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

There are 4 distinct launch sites in the data

Launch Site Names Begin with 'CCA'

%sql SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

5 records are shown, where launch sites begin with ‘CCA’

Total Payload Mass

%sql SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Customer LIKE "%NASA (CRS)%";

SUM(PAYLOAD_MASS__KG_)
48213

The total payload mass carried by boosters launched by NASA is 48213 kg

Average Payload Mass by F9 v1.1

%sql SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTABLE
WHERE Booster_Version LIKE "%F9 v1.1%";

AVG(PAYLOAD_MASS__KG_)
2534.6666666666665

The average payload mass carried by booster version F9 v1.1 is 2534.67 kg

First Successful Ground Landing Date

%sql SELECT Date FROM SPACEXTABLE WHERE Landing_Outcome LIKE "Success (ground pad)" ORDER BY Date ASC LIMIT 1;

Date
2015-12-22

The first successful landing outcome in ground pad was achieved on Dezember 22 in 2015

Successful Drone Ship Landing with Payload between 4000 and 6000

%sql SELECT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome == "Success (drone ship)" AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000;

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

The listed boosters have a payload between 4000 and 6000 kg and had successful drone ship landings

Total Number of Successful and Failure Mission Outcomes

%sql SELECT COUNT(*) FROM SPACEXTABLE WHERE Landing_Outcome IN ('Failure (parachute)', 'Failure (drone ship)', 'Success (ground pad)', 'Success (drone ship)', 'Success', 'Failure');

COUNT(*)
71

A total of 71 successful and failure mission outcomes were registered

Boosters Carried Maximum Payload

%sql SELECT DISTINCT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ == (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);

All boosters that carried maximum payload are listed

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

%sql SELECT substr(Date, 6,2) AS 'Month', Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTABLE WHERE substr(Date,0,5)='2015';

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
02	Controlled (ocean)	F9 v1.1 B1013	CCAFS LC-40
03	No attempt	F9 v1.1 B1014	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40
04	No attempt	F9 v1.1 B1016	CCAFS LC-40
06	Precluded (drone ship)	F9 v1.1 B1018	CCAFS LC-40
12	Success (ground pad)	F9 FT B1019	CCAFS LC-40

A selection of information for 2015 launches are shown sorted by month

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

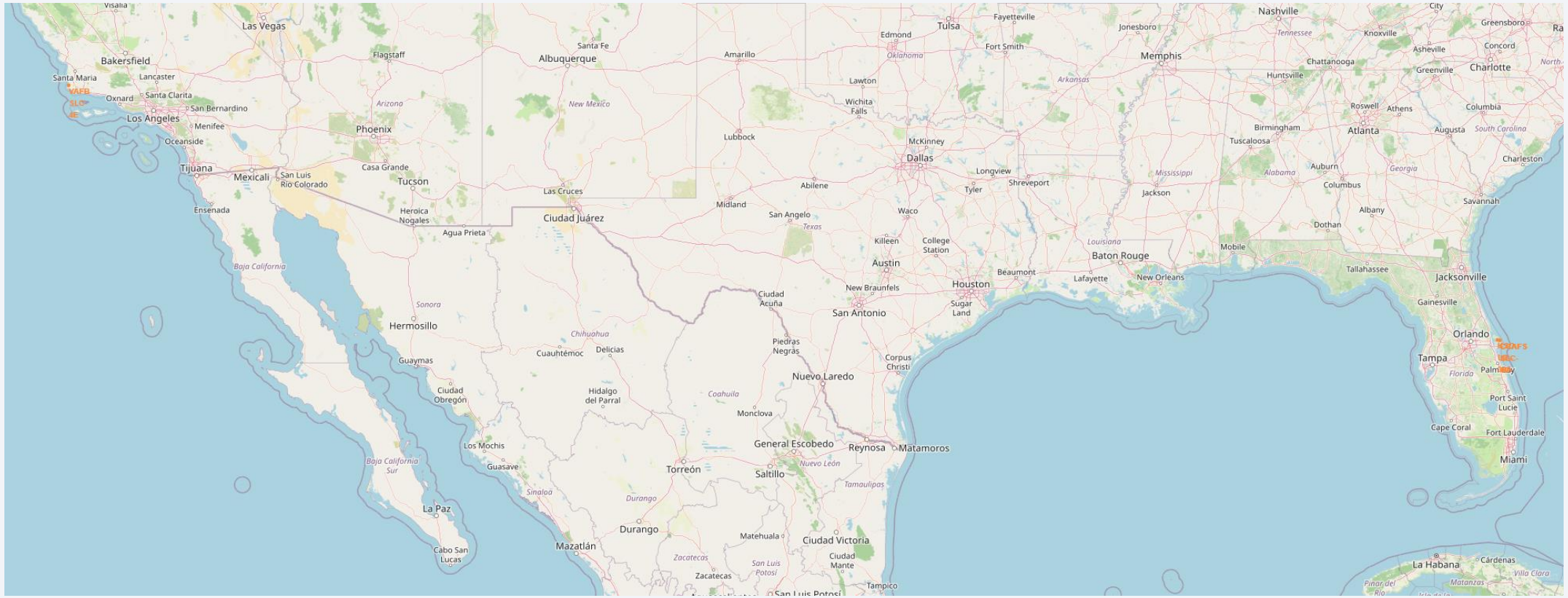
%sql SELECT Landing_Outcome, Count(*) FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Count(*) DESC;

The count of landing outcomes between 2010-06-04 and 2017-03-20 is shown in descending order

Landing_Outcome	Count(*)
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1


Section 3
Launch Sites
Proximities Analysis

Launch site locations



All launch sites are located close to the coastline, 1 at the west coast, 2 at the east coast of the USA

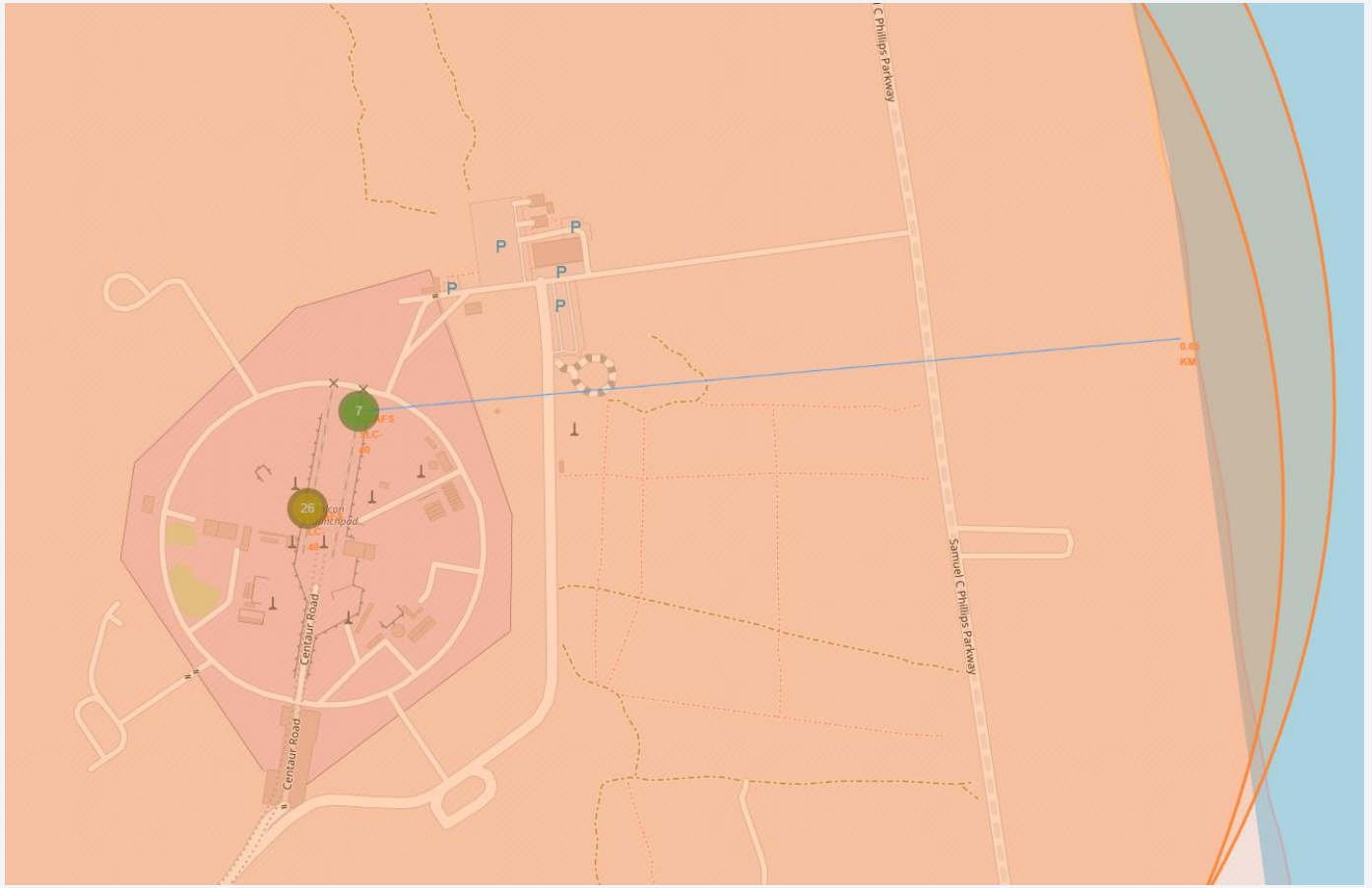
Launch outcomes

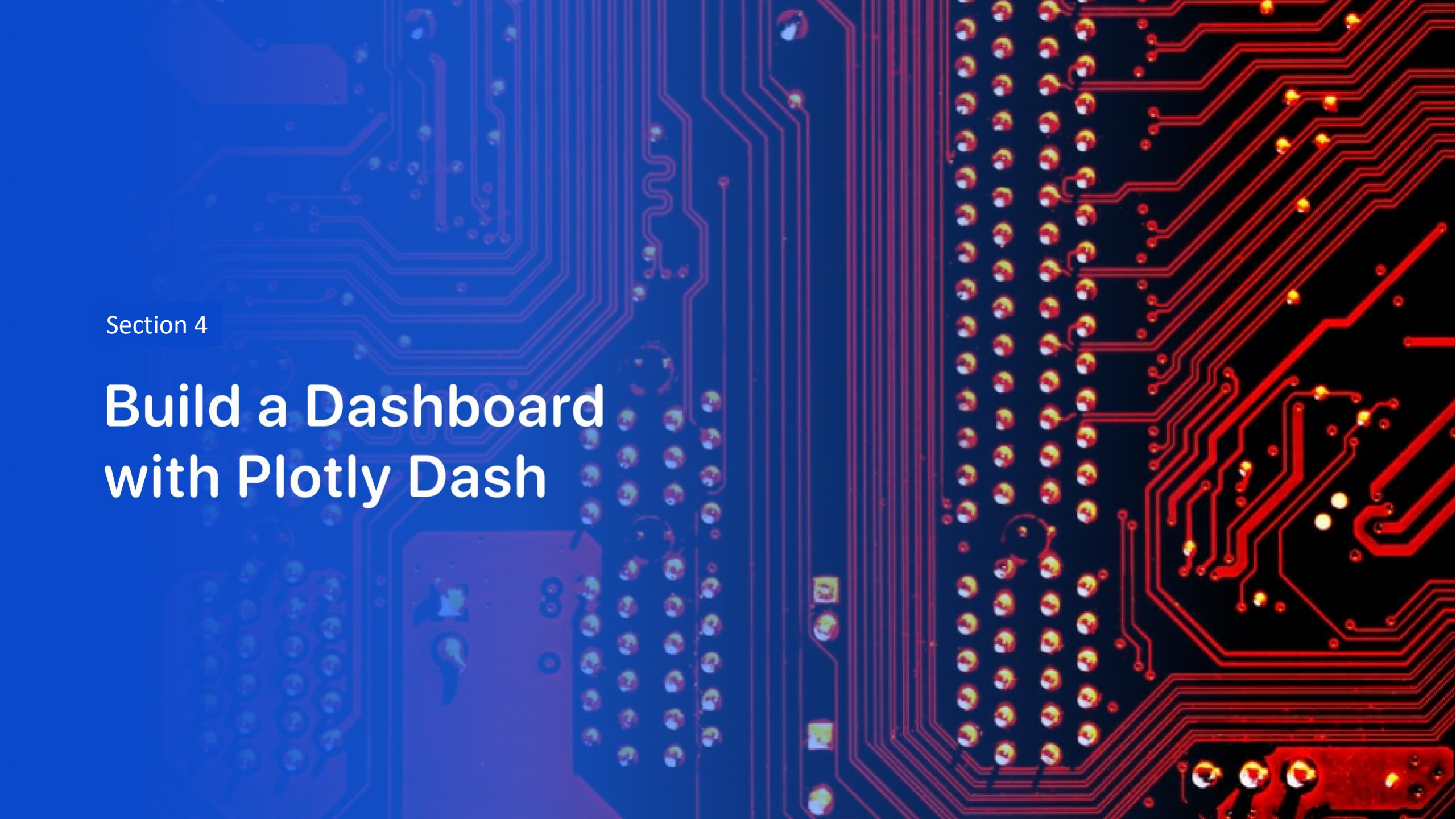


It can be seen that the selected launch site has a relatively high success rate

<Folium Map Screenshot 3>

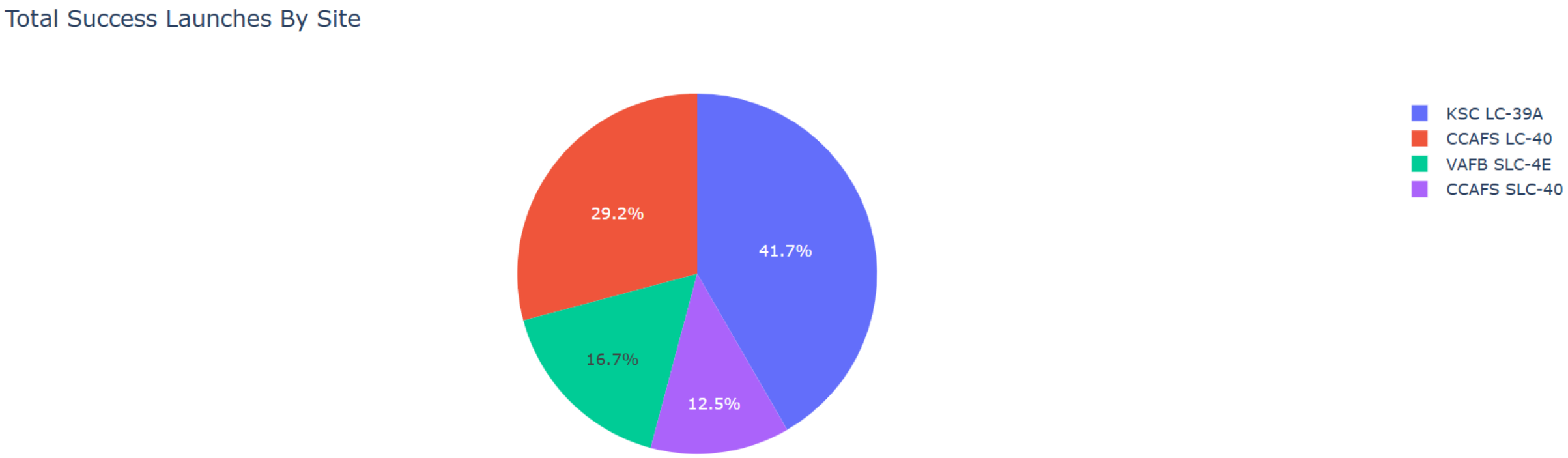
It can be seen that the selected launch site is located 0.85 km from the coastline.




Section 4
Build a Dashboard with Plotly Dash

SpaceX Launch Records Dashboard

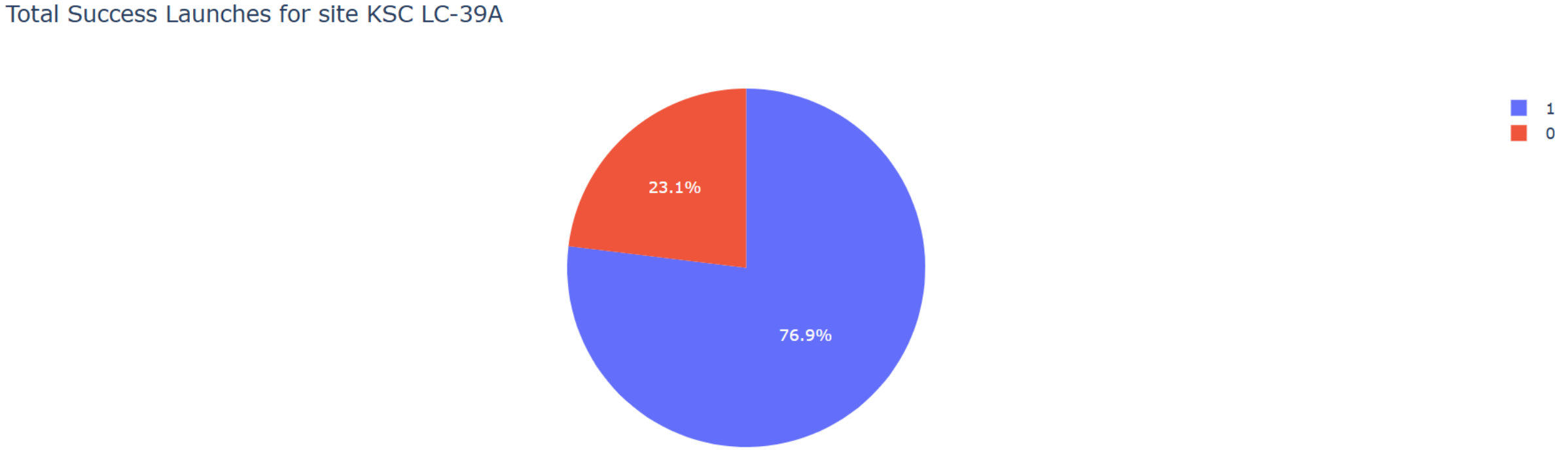
All Sites



- The highest number of successful launches were at site KSC LC-39A
- The lowest number of successful launches were at site CCAFS SLC-40

SpaceX Launch Records Dashboard

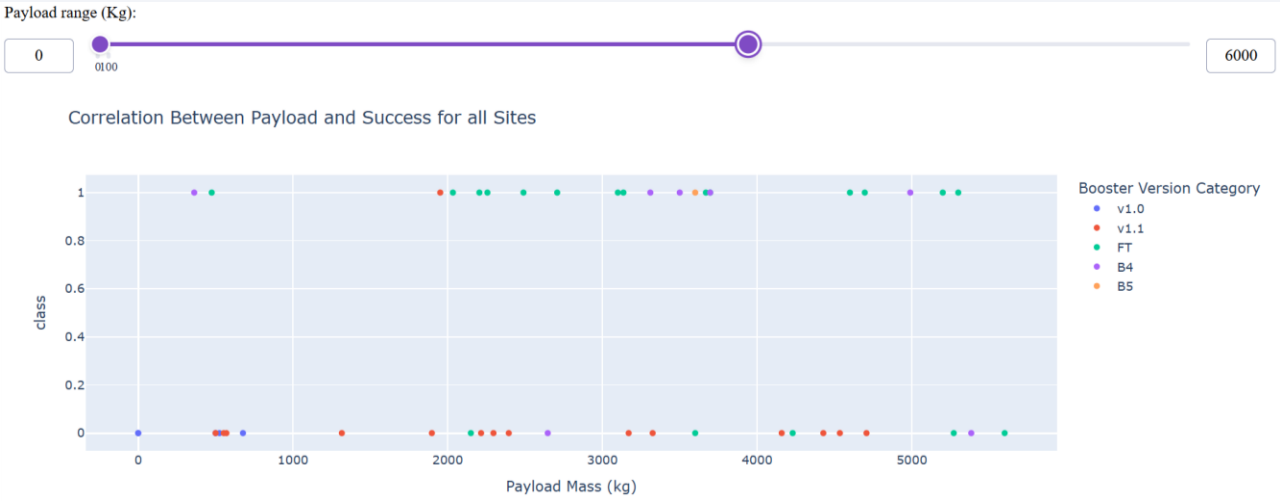
KSC LC-39A



- 76.9 % of launches were successful at KSC LC-39A
- Close to ¼ of the launces were unsuccessful

Correlation between Payload and Success







- Booster versions 1.0 and 1.1 have very low success rates and have less than 5000 kg payload
- Booster versions FT and B4 have more successful landings than unsuccessful, while having both low and high payloads
- Payloads between 5000 kg and 7000 kg have very low success rate

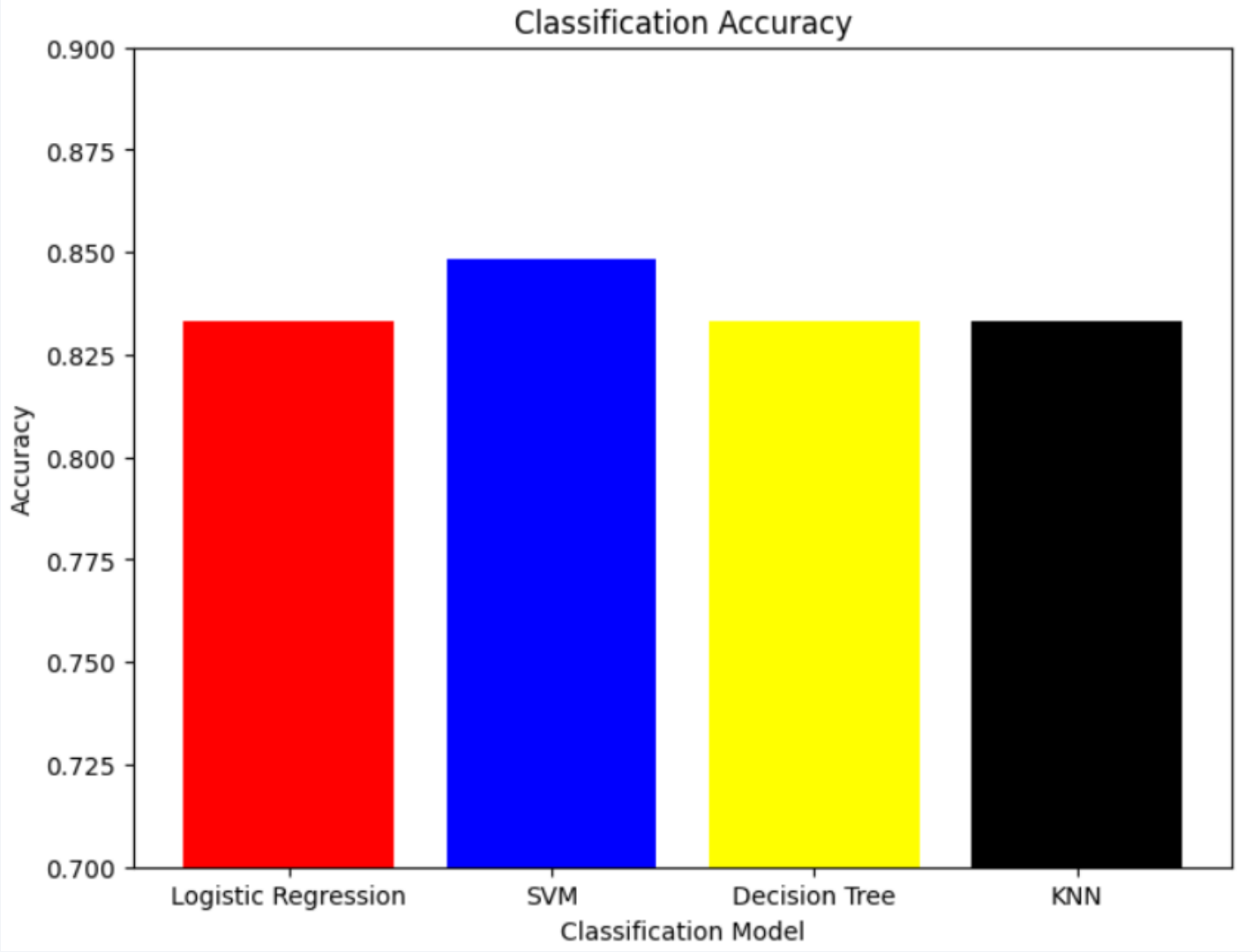
Section 5

Predictive Analysis (Classification)



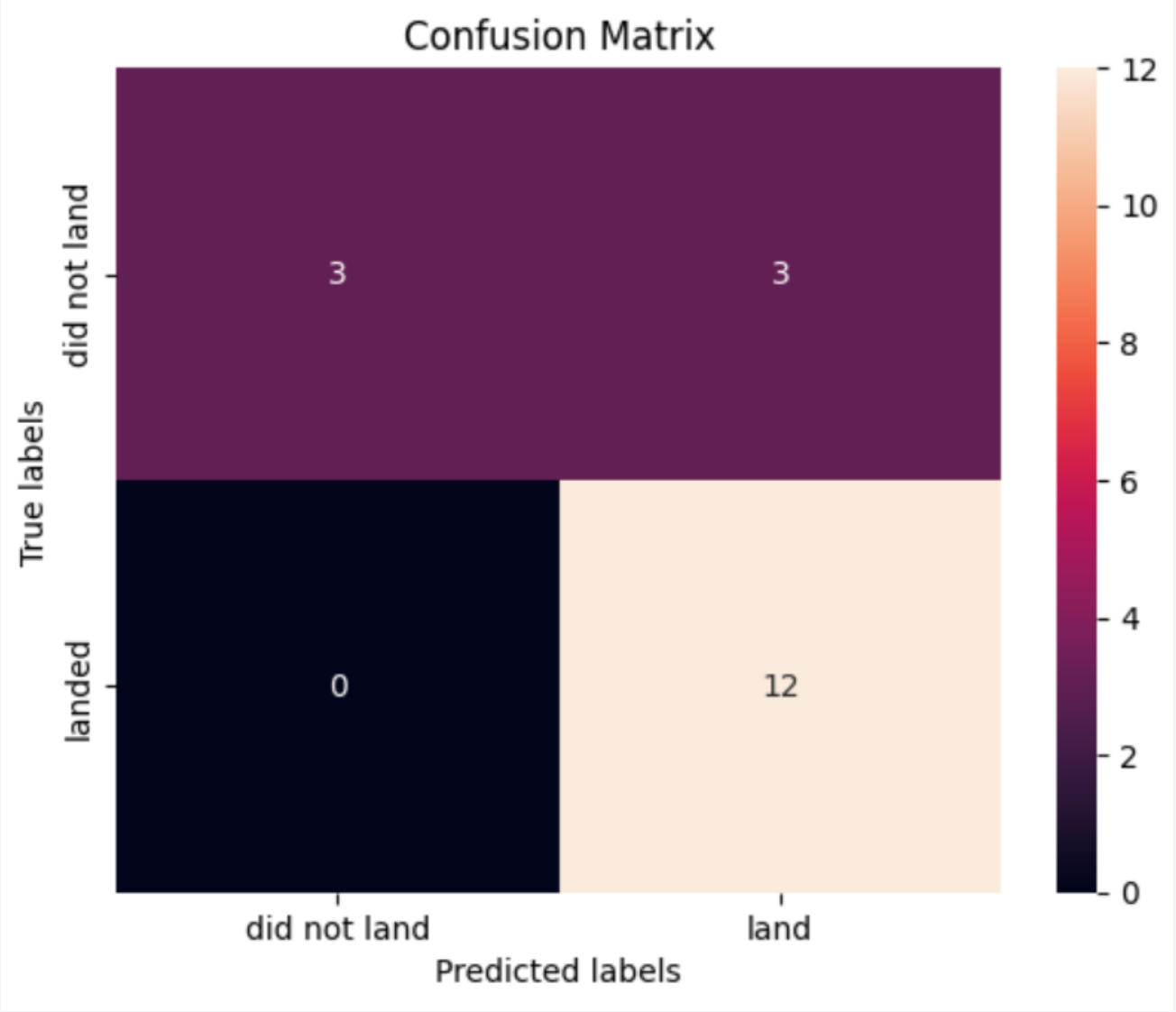
Classification Accuracy

SVM has the highest classification accuracy with 84.8 %



Confusion Matrix

No false negatives and 3 false positives were predicted by SVM.



Conclusions

- SpaceX is increasingly successful in landing the first stage and is able to increase payload mass, while maintaining high success rates.
- For high payloads especially orbit type VLEO is highly successful
- Multiple launch sites are being used, with KSC LC-39A having the most successful landings.
- Launch sites are located close to the coastline and with access to infrastructure, while keeping distance to cities
- Booster versions FT, B4 and B5 show the most promising success rates and can facilitate a wide range of payload mass
- Different classification models for successful landings were created with SVM performing best (84.8% accuracy). While it accurately predicts false negatives, false positives might be predicted at a significant rate, leading to an overall overestimation of successful landings

Thank you!

